Department of Physics
Ariel University, Israel
+972 55 263 1094
meamresh.github.io
Citizenship: India

Amresh Verma

Research Profile

PhD candidate in theoretical cosmology and gravitational-wave phenomenology, with six refereed journal papers, a conference proceedings and one manuscript under review, spanning stochastic gravitational-wave backgrounds, pulsar timing array (PTA) analysis, early-universe physics, and CMB / large-scale-structure anomalies. Active research lines include PTA anisotropy systematics, primordial black-hole-driven gravitational-wave signatures, and Bayesian inference under model misspecification. Methodological strengths in MCMC, sequential Monte Carlo, Fisher analysis, differentiable inference, and probabilistic machine learning, with hands-on experience in JAX, HEALPix, FastPTA, PTArcade, CAMB, and CLASS.

Education

2023–2027 (expected) **Ph.D. in Physics**, *Ariel University*, Israel
Thesis: *Phenomenological Aspects of Cosmology*
Advisor: Prof. Ido Ben-Dayan
Defence late 2026; completion expected January 2027

2020–2022 **M.Sc. in Physics (Distinction)**, *Jamia Millia Islamia*, New Delhi, India
CGPA: 9.26/10
Thesis: *Study of Dark Matter–Dark Energy Interaction in the Light of Hubble Tension*
Advisor: Dr. Somasri Sen

Publications

1. Ben-Dayan, I., Kumar, U., Thattarampilly, U., **Verma, A.** (2023). *Probing the Early Universe Cosmology with NANOGrav: Possibilities and Limitations*. Phys. Rev. D 108, 103507.
2. Ben-Dayan, I., Calcagni, G., *et al.*, **Verma, A.** (2024). *Gravitational-wave background in bouncing models from semi-classical, quantum and string gravity*. JCAP 09 (2024) 058.
3. Ben-Dayan, I., Kumar, U., Shimon, M., **Verma, A.** (2025). *Impact of Low-ℓ's on Large Scale Structure Anomalies*. JCAP 02 (2025) 069.
4. Ben-Dayan, I., Kumar, U., **Verma, A.** (2026). *Disentangling the Origins of the NANOGrav Signal: Early Universe Models and $\Delta N_{\rm eff}$ Bounds*. JHEAp 50 (2026) 100510. *(Designed and led the analysis; handled co-author coordination and correspondence with the editor and referees.)*
5. Kumar, U., Ajith, A., **Verma, A.** (2026). *Cosmological constraints on non-cold dark matter from DESI DR2 measurements*. Phys. Lett. B 880, 140858.

6. Ben-Dayan, I., Kumar, U., **Verma, A.** (2026). *The PTA–CMB "tension" and bounds on extra radiation from Early-Universe models*. PoS CORFU2025 (2026) 227.
7. Akama, S., *et al.* [incl. **Verma, A.**] (2026). *Testing gravitational wave polarizations with LISA*. JCAP 09 (2026) 055.
8. Ben-Dayan, I., Srivastava, A., **Verma, A.** (2026). *Slowly Rolling on a Quantum Correction*. arXiv:2608.25004 [astro-ph.CO] (in peer review).

Research Experience

Sep–Nov 2025 **Research Visit**, *Deutsches Elektronen-Synchrotron (DESY)*, Hamburg, Germany
Host: Dr. Thomas Konstandin. Worked on systematics in PTA anisotropy searches: extended the public FastPTA framework (Domcke, Franciolini & Pieroni 2025) with a white-noise sensitivity study and derived a custom frequency-leakage window function.

Selected Research Projects

A. **Geometry-Induced Monopole Bias in Pulsar Timing Arrays.** Solo study showing that incomplete pulsar sky coverage turns gravitational-wave anisotropy into a systematic monopole bias. Quantified the distribution of the bias-to-error ratio B by covariance traces, exact for the specified discrete covariance and validated against an audit of 810 Monte Carlo batches. Isotropic catalogs give a spread $\mathrm{SD}(B) \simeq 0.62$ at $C_1/C_0 = 0.09$, nearly independent of array size, with $|B| \geq 1$ in about 11% of configurations; restricting pulsars to declination above -40° raises the spread from 0.79 at 20 pulsars to 2.30 at 400. The 67 public NANOGrav positions give $\mathrm{SD}(B) = 2.008$ under the same idealized assumptions. JAX/HEALPix forward model. [In preparation] [GitHub]

B. **Frequency-Leakage Window Function for PTA Anisotropy Searches.** Extension of the public FastPTA framework (Domcke, Franciolini & Pieroni 2025). Derived a custom finite-window window function and demonstrated, via a controlled null injection test, that it suppresses the spurious cross-band dipole bias by ∼40% at matched $N_p = 300$, 400, 500: at $N_p = 500$, standard windowing produces a 2.07σ deviation from null while the new window keeps it at 1.25σ. Conducted in part during the DESY research visit. [GitHub]

C. **Multi-Species Primordial Black-Hole Cascade and Induced GW Comb.** End-to-end forecasting pipeline for cosmologies with N PBH mass species undergoing sequential matter-to-radiation transitions, with constraint coverage (ΔN_{eff}, FIRAS μ/y, BBN entropy). Investigating whether the cascade produces a structured N-peak induced gravitational-wave spectrum separable from single-species scenarios, and extending the treatment from monochromatic to extended mass functions. Detectability forecasts across the PTA, LISA and ET bands in progress. [In preparation]

Conference Presentations

2025 Oral Presentation, *Synergies Towards the Future Standard Model*, DESY, Hamburg.

2025 Flash Talk & Poster, *High Energy and the Cosmos Workshop*, AGASS, Ariel University.

2024 Poster, *CosmoVerse@Krakow 2024*.

2024 Oral Presentation, *69th Annual Meeting of the Israel Physical Society*.

2023 Oral Presentation, *Current Trends in Cosmology and High Energy Physics*, CCSP.

Teaching & Outreach

Teaching Alpha Programme (Israel), a national research programme for high-school students: taught participants to fit ΛCDM to supernova data and estimate H_0, covering introductory programming, Bayesian methods and cosmology.

Mentoring Trained a co-author in data-analysis methods through to publication. Tutored high-school students in physics in my home village in India.

Outreach Run a public science-communication channel on cosmology and scientific scepticism.

Schools & Workshops

2025 **CosmoVerse Workshop @ Naples**, Naples, Italy.

2025 **LULBI – Light & Ultralight BSM Interactions**, Weizmann Institute of Science, Israel.

2024 **SIGRAV International School – Measuring Gravity**, Italy.

2022 **Machine Learning in Solar Physics and Space Weather**, IISER Kolkata.

2022 **IASTRO Summer Internship**, Instituto de Astrofísica e Ciências do Espaço, Lisbon.

Awards & Grants

2025 AGASS Excellence Scholarship in Space Research and Astro-Informatics.

2024 COST Grant for presenting research at CosmoVerse@Krakow 2024.

Technical Skills

Cosmology Software CAMB, CLASS, MGCAMB, PTArcade, Cobaya, FastPTA, HEALPix.

Bayesian Inference MCMC (HMC, NUTS, PMMH), sequential Monte Carlo, particle filters (bootstrap, differentiable, Sinkhorn-OT resampling), Fisher information analysis, posterior calibration diagnostics.

Probabilistic ML Bayesian neural networks, variational inference, Hamiltonian neural networks, uncertainty quantification (aleatoric vs. epistemic), model-misspecification analysis.

Programming Python (SciPy, JAX, PyTorch, TensorFlow, Numba, pandas), C, Fortran, Git.

References

Available on request.